

Vansh Parikh

vansh.parikh@mail.utoronto.ca — GitHub: VanshParikh1 — LinkedIn: vanshparikh22

Education

University of Toronto — Mississauga

Expected Graduation: November 2027

Bachelor of Science in Statistics Minors: Computer Science, Business, Science and Entrepreneurship (IMI Program)

Experience

Microsoft Power Platform Mentorship

May 2026 – June 2026

Copilot Studio, Power Automate, Dataverse, SharePoint, Power BI

- Designed and shipped an end-to-end HR leave management system: Copilot Studio agents, Power Automate approval workflows, and Dataverse for structured employee data modeling.
- Built multi-step approval pipelines with event-driven triggers and conditional logic, and connected Copilot Studio agents to SharePoint as a live knowledge source for document-driven retrieval.
- Owned the end-to-end solution architecture across Copilot Studio, Dataverse, Power Automate, SharePoint, and Power BI reporting.

African Geospace

September 2025 – December 2025

Mobile Developer — React Native, Expo, Flask

- Stabilized and upgraded the React Native/Expo development environment by resolving SDK and dependency conflicts, enabling reliable cross-device testing and deployment.
- Built core mobile features including navigation architecture, category-based browsing, and reusable UI components to improve maintainability and scalability.
- Integrated frontend systems with backend APIs, handling authentication, data flow, and debugging cross-system issues in a full-stack environment.
- Communicated directly with non-technical stakeholders, translating technical progress into clear updates and adjusting priorities based on client feedback.

Projects

SketchQuest – AI Sketch-to-Playable-Game Pipeline

1st Place, EmberHacks 2026 (84 participants)

React, TypeScript, Vite, Phaser 3, Node.js, Express, Gemini Vision API, Zod

GitHub Repository

- Led a 3-person team to build an end-to-end pipeline converting a photographed hand-drawn platformer level into a playable Phaser 3 game, using Gemini Vision to interpret the sketch into structured JSON level data.
- Designed a reachability validation and auto-repair loop that replays the game's own physics assumptions to detect unbeatable levels and has Gemini minimally repair them before load, preventing broken gameplay from reaching the player.
- Engineered a resilience layer (response caching, mock/fallback data, request-rate limiting) around the Gemini API to keep the app fully demoable under latency and quota constraints.

Zoink – Peer-to-Peer Student Rental Marketplace

March 2026 – Present

React Native, Node.js, Express, PostgreSQL, Prisma, TypeScript

GitHub Repository

- Architected a full-stack monorepo (React Native/Expo, Node.js/Express, PostgreSQL via Prisma, shared TypeScript DTOs) with secure RESTful APIs for auth, listings, image uploads, and booking state management, powering real-time listings, bookings, and in-app messaging.
- Redesigned the booking-conflict check into a sorted-interval algorithm with $O(\log n)$ binary-search lookups and added a missing composite database index, eliminating a full-table scan on every booking accept/confirm; shipped with 31 new unit tests and zero regressions across a 362-test suite.
- Designed and shipped a tiered commission and payout engine (rate tiers by daily listing price) with automated deposit capture/release tracking and HST handling, integrated with Stripe Connect for escrow and payouts.

Whiskr – AI-Powered Cat Wellness iOS Application

May 2025 – August 2025

Swift, SwiftUI, AVFoundation, UserNotifications, OpenAI API

GitHub Repository

- Designed and built a 3-tab MVVM iOS app solo (30+ SwiftUI views/models, a 7-step onboarding flow, and Codable-based local persistence) to help owners manage cat wellness and daily care routines.
- Integrated OpenAI's Responses API (GPT-5-mini) directly over `URLSession` with typed `Codable` response/error parsing and dynamic system-prompt injection of live cat profile data, powering a personalized AI symptom-checker chatbot.

Technical Skills

Languages: Python, TypeScript, Swift, Java, C, R, SQL

Mobile: React Native, Expo, SwiftUI, React Navigation, AVFoundation, Xcode, Android Studio

Backend & Data: Node.js, Express, Flask, PostgreSQL, Prisma ORM, REST APIs, Stripe

AI/ML & Tools: Gemini API, OpenAI API, React, Phaser, Git, UNIX/Linux, VS Code, Xcode